

Unit 2

Demand & Supply :-

Demand analysis - Demand is important concept in macro economics the demand for any commodity is the relationship between the price & the quantity that will be purchased at that price. In other words, the demand for a good is a schedule of the amount that ~~demanded~~ ~~for a~~ they would be willing to purchase at possible price according to a specific time. The demand for any thing at a given price is the amount of it which will be bought per unit of time at that price.

⇒ According to above definition demand has some important essential.

- (1) To have an effective desire & ability to fulfill the desire & willingness to use those resources for fulfillment of their desire. That means we need money to purchase the product.
- (2) Demand for a particular commodity is related with a particular price.

(3) Demand for a particular commodity is selected with a particular time.

Examples:-

- (1) Demand for commodity 'A' is 1000 units.
- (2) Demand for commodity 'B' at 5 rs ~~per~~ unit.
is 1000 units.
- (3) Demand for commodity 'C' at 5 rs per unit in Jaipur is 1000 unit/week.

* Characteristics:-

- Demand is an effective desire.
- Demand is related with price.
- Demand is related with time.

* Importance:-

- Importance in consumption
- Importance in exchange
- Importance in distribution.
- Importance in Finance.
- Advantage to procedure
- Importance in religion & culture or politics.

* Types of demand:-

- Income demand:- It is related with income. It is now the relationship between the quantity demand of a commodity & the

income of consumer.

Income demand implies the quantity of commodity purchased by the consumer at various level of income. otherwise that is taste, fashion. don't, change.

- Price demand:- It implies the different quantity of a commodity demanded by consumer at various price for a particular period of time that means income, taste, fashion is unchanged.
- Cross demand:- It is related to sub-situational product this is related with substitution & complementary goods.

Determinants of demand:-

(Factor in ~~Factor~~ Hunting demand)

- (1) Number of consumer.
- (2) Income level
- (3) Distribution of wealth
- (4) ~~Government~~ government control.
- (5) Consumer taste, fashion, preference etc.
- (6) Business condition.
- (7) Improvement in the quality.
- (8) Changes in population.
- (9) Changes in money supply.
- (10) ~~Substitution~~ Substitution available in the market.

* Demand Forecasting :-

Demand Forecasting is related with the concept of demand. Demand forecasting refers to the ~~product~~ prediction of ~~dem~~ estimated demand for the quantity to be produced for the market. In other words we can say that it is an estimation of future demand. According to Kotler, demand forecasting as a market demand for a ~~production~~ is the total volume that would be what, why or define customer ~~group~~ growth in a defined geographical area, in a defined time period. Business is always depends on forecasting. And it is the important method production & manufacturing as well.

Objectives of demand Forecasting :-

(Objective of
DF)

Short term objective

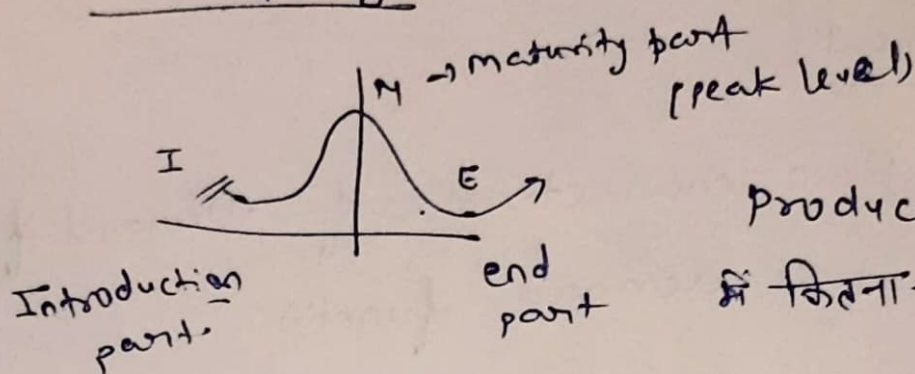
- Better financial planning
- Determining pricing strategy.
- Better financial management.
- Formulation of sales strategy.
- Proper production schedule.

Long term objective

- long term financial planning.
- Project expansion
- Development of forward linkage to backward linkage.
- Adoption of new strategy or technology.

Improving
Product
life
cycle.

Product life cycle.



* Steps involve in demand forecasting :-

Step 1. Identification of production



Step 2. Determining objectives of demand forecasting.



Step 3. Determining scope of demand forecasting.



Step 4. Identifying the risk involve in demand forecasting.



Step 5. Collection & analysis of data.



Step 6. methodology is to be used in DF.



Step 7. Interpretation Conclusion & decision.



Step 8. reviewing the decision.



Step 9. Result & feedback.

Assignment

Q1 Explain characteristics of Demand forecasting.

Q2 What is demand function.

Q3 Explain elasticity of demand.

Q4

* Methods of Demand Forecasting :-

There are 6 methods of demand forecasting.

(1) Survey method :-

- (a) Survey of buyer's intention.
- (b) Survey of sales-force.
- (c) Survey of expert opinion.
- (d) Survey of opinion poll-survey.

(2) Study & experiments :-

- (a) Test marketing
- (b) Experiments

(3) End-user methods :-

- (a) Listing of all present & future, possible use of product.
- (b) Designing and technical norms of product.
- (c) Stages related to the Application of product.
- (d) Stage related with product-wise use & user-wise and use wise.

14) Statistical method & analysis :-

(a) brand projection method. (Graphical method)

(b) Regression method.

(c)

(5) Baro-metric method of Forecasting :-

(a) Leading indicator - new orders, corporate profit & value of inventory.

(b) Coincidental indicators - level of economic activity, rate of unemployment constant-price & sells record by manufacturing

(c) Lagging indicators - ~~long~~ loan outstanding & sells return.

(6) Econometric method :-

Related with data; regression equations and variables.

* Problems in demand forecasting :-

(1) Government rules & regulation

(2) Government control.

(3) Incapability of management decision

(4) Insufficient funds.

(5) Poor technology.

(6) Lack of demand of product

Good characteristics of demand forecasting -

- (1) Accuracy
- (2) Better financial planning
- (3) Simplicity.
- (4) Good managerial decision
- (5) Flexibility
- (6) Economy-wise good.

* Supply

Supply is a flow related with per day, per month or per year.

Supply is the amount offered for sale per unit of time.

There ~~are~~ is two parties in supply, i
(i) Saler (ii) Producer.

Supply is the total quantity offer commodity which a saler would be willing to produce & sell at a give price during a specific time. It is also related with the different type of situation.

Factor Affecting supply -

- (1) Price of the commodity
- (2) Price of factor of production.
- (3) Price of related commodity.
- (4) State of technology.

- (5) Government policy.
- (6) Operational law of returns.
- (7) Time period.

Assignment Question

Q3 Explain characteristic of demand along with demand function & elasticity of demand.

Q4 Explain determinants of supply along with supply function & elasticity of supply

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Q1 Explain characteristic of demand forecasting.
Ans In economics, demand forecasting refers to predicting the future demand for goods & services in a market for economy. Here are the key characteristics in the economics context:-

Characteristics:-

- (1) Futuristic in Nature:- Forecasting is always forward-looking. It estimates future demand for goods / services to help producers, policymakers & investors.
- (2) Uncertainty involved:- Future demand depends on many uncertain factors like consumer tastes, income levels, population growth & government policies.
- (3) Department on time horizon:- Forecasting may be short-term (month to a year e.g. inventory decisions) or long-term (5-10 years, ex. capacity expansion in industries).

Based on Assumptions:- Economics forecasting assumes stable conditions of factors like prices, income, technology & competition.

Use of Historical Data:- Past sales, consumption pattern & economic indicators are used to estimate future demand trends.

- (6) Quantitative & Qualitative Approaches:-
Economics used both statistical models (e.g. regression, trend analysis) & qualitative judgements (e.g. expert opinions, Delphi methods).
- (7) Dynamic Process:- Since economic conditions change (inflation, policy shifts, global trends), forecasts must be updated frequently.
- (8) Goal-oriented:- The main aim is to assist in economic planning, pricing policies, resource allocation & production strategies.
- (9) Influenced by External Factors:- Economic demand forecasts depends on population, income distribution, employment levels, government regulation & international trade.
- (10) Helps in Policy-making:- Governments & businesses use demand forecasting to frame monetary policy, fiscal policy, taxation & industrial strategies.



What is demand function?

The demand function shows the relationship between the quantity of a product demanded and the factor affecting it, mainly its price, consumer income, tastes & the prices of related goods.

General form

$$Q_d = f(P, Y, P_r, T, E)$$

where

Q_d = Quantity demanded

P = Price of the product

Y = income of consumers.

P_r = Prices of related goods

T = Tastes & preference.

E = Expt. Expectation about future

as linear function $Q = a - bP$ (changes)

$$Q = a - bP$$

It helps managers forecast sales, set prices & plan production

Q3 Explain elasticity of Demand?

Ans Elasticity of demand measures the responsiveness (sensitivity) of quantity demanded of a good or services when there is a change in any of its determinant, mainly price.

Types of elasticity of demand

a) Price elasticity of demand -

$$E_p = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

(2) Cross elasticity of Demand -

$$E_c = \frac{\% \text{ Change in quantity demanded of good } x}{\% \text{ change in Income}}$$

Advertisement

(3) ~~Cross~~ elasticity of Demand -

$$E_a = \% \text{ change in quantity demanded of good}$$

- Shows responsiveness of demand to change in advertising or promotional expenditure.
- Degree of price elasticity.

(1.) Perfectly elastic (∞)

(2.) Elastic (>1)

(3.) Unitary ($=1$)

(4.) Inelastic (<1)

(5.) Perfectly Inelastic (0)

(4) Ex

Q4 Explain determinants of supply along with supply function & elasticity of supply.

Ans Supply means the quantity of a good or services a producer is willing and able to sell at different prices in a given period of time.

(1) Price of the commodity (P_x)

(2) Price of related goods (P_r)

(3) Cost of production (C)

(4) Technology (T)

(5) Number of sellers.

(6) Government Policy (G)

(7) Expectations of future prices (E)

(8) Natural & climatic condition (C_1)

Supply function :- The supply function shows the functional relationship between the quantity supplied & the factor affecting it.

General Form -

$$Q_s = f(P_x, P_r, C, T, N, G, E, Cl)$$

Q_s = Quantity supplied

P_x = Price of the good

P_r = Price of related good

C = Cost of production

T = Technology

N = Number of sellers

G = Government Policy

E = Expectations

Cl = Climate conditions.

$$Q_s = f(P_x)$$

Elasticity of Supply :- Elasticity of supply measures the responsiveness of quantity supplied to a change in price.

$$E_s = \frac{\% \text{ change in Quantity Supplied}}{\% \text{ change in price}}$$

1. Perfectly elastic supply ($E_s = \infty$)
2. Elastic supply ($E_s > 1$)
3. Unitary elastic supply ($E_s = 1$)
4. Inelastic supply ($E_s < 1$)
5. Perfectly inelastic supply ($E_s = 0$)